Link to the OneDrive:

https://xmueducn-my.sharepoint.com/:f:/g/personal/ait2509090_xmu_edu_my/IgCD_2YKeahiSZj9srjngDQIAYE3O66bpqt-YRdklWRXDK8

How to Run the Application

Prerequisites

Ensure Python 3.12 is installed on your machine along with all required dependencies. Navigate to the project folder and run the following command:

pip install -r requirements.txt

Launching the Application

Once dependencies are installed, launch the application with:

python app.py

No additional setup is required. The Cambridge road network graph and the trained triage model are pre-bundled in the data_files/ folder, so the application starts immediately without any downloads.

Using the System

1. On the **Dashboard**, select a call volume from the dropdown (10, 20, or 30 concurrent emergency calls)
2. Click **Generate Scenario** to create a randomised emergency scenario
3. Click **Run AI** to execute the full four-stage pipeline
4. Navigate to the **Interactive Map** page to view ambulance routes, hospital assignments, and emergency call locations on the Cambridge road network
5. Navigate to the **Analytics** page to view the confusion matrix, assignment comparison, routing comparison, and hospital recommendation results

Note: An active internet connection is required for the Interactive Map page to display OpenStreetMap tiles. The AI pipeline itself runs fully offline using the locally cached road network.